



GLOBAL ECONOMICS
GRADE: 11TH
TERM 3 — LEARNING EVIDENCE 4
WEB IMPLEMENTATION OF HYPOTHESIS
TESTING (C8–C9)
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THIRD
TERM
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Learning objective: Build a functional webpage that extends the C1–C7 hypothesis-testing process into an interactive digital tool, including manual parameter control, visual p-value interpretation on a normal curve, and automatic analysis from uploaded two-group CSV data.

Assessed criteria:

C8: Creates an interactive webpage that represents the p-value of a hypothesis test on a normal curve and allows the user to modify the test parameters.

C9: Creates a webpage that imports two-group CSV data and automatically performs a hypothesis test using the uploaded data.

Background from C1–C7

In C1–C7 you already practiced the full statistical reasoning cycle: writing H_0 and H_1 , selecting the tail direction, calculating test statistics, making decisions with α , interpreting errors, and determining minimum sample sizes. In this activity, you will transform that reasoning into a working webpage.

Task overview

Create one webpage (HTML + CSS + JavaScript) that performs hypothesis testing in two connected modes:

1. **Manual mode (C8):** users set parameters and visualize the p-value on an interactive normal curve.
2. **CSV mode (C9):** users upload two-group data and the webpage computes the hypothesis test automatically.

The page must be functional, not only visual. Do not submit worked solutions, completed answer sheets, or screenshots as a replacement for functionality.

Mandatory requirements for C8

Your C8 implementation must include all of the following:

- Input controls using both **sliders** and **input boxes**.
- Test-direction selector: left-tailed, right-tailed, two-tailed.
- Significance level α .
- Enough parameters to calculate z .
- Null value (Hypothesis).
- Sample statistic.
- Standard deviation or standard error.
- Sample size.

The output panel/graph must show:

- The normal curve.

- The observed test statistic.
- The shaded p-value region.
- The numerical p-value.
- The critical value or critical region.
- The statistical decision: reject H_0 or fail to reject H_0 .
- A short contextual conclusion consistent with the decision.

Mandatory requirements for C9

Your C9 implementation must include all of the following:

- CSV upload control.
- Reading data from two groups.
- Detection of group columns, or user selection of group columns.
- Computation of summary statistics for each group.
- Selection of benchmark group and test group.
- Test-direction selector: left-tailed, right-tailed, two-tailed.
- Automatic execution of the hypothesis test.
- Display of test statistic, p-value, decision, and contextual conclusion.
- Display of the p-value on the same type of normal-curve graph used in C8.

Required CSV example files

Use and reference these attached files:

- G11_T3_L4_C8C9_example_mean_data.csv
- G11_T3_L4_C8C9_example_proportion_data.csv

Expected structure:

- **Mean-based file:** two group columns containing numerical observations (for example counts, times, or scores).
- **Proportion-based file:** two group identifiers plus a binary success variable (0/1) for each observation.

These two files must run correctly in your webpage and be used to validate your implementation.

Deliverables

Submit all of the following:

1. `index.html` (or equivalent main HTML file).
2. Linked stylesheet file(s) (CSS).
3. JavaScript file(s) with calculation and graph logic.
4. The two attached CSV files.

5. A short README describing:

- how to open the webpage,
- how to test C8 mode,
- how to test C9 mode with both CSV examples.

Evaluation checklist (validation requirements)

Your work is valid only if all checks pass:

1. The graph updates when parameters change.
2. The p-value updates when parameters change.
3. The decision updates when parameters change.
4. The CSV files are read correctly.
5. The test is computed correctly from uploaded data.
6. Both attached CSV examples work in the webpage.
7. The contextual conclusion matches the displayed decision.

Submission instructions

Before submitting, verify this checklist:

- ☐ I implemented HTML structure, CSS styling, and JavaScript logic.
- ☐ I included sliders and input boxes for manual parameter control.
- ☐ I implemented CSV upload and automatic hypothesis-test calculation.
- ☐ I display the normal curve, p-value region, p-value, decision, and conclusion.
- ☐ I tested the webpage with both attached CSV example files.
- ☐ My final conclusion is consistent with the statistical decision.